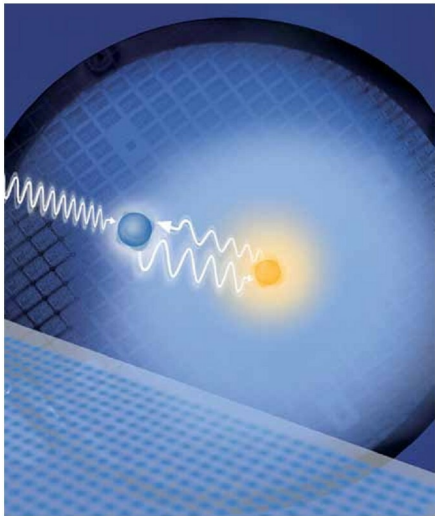


COMPUTING TAKES QUANTUM LEAP

While molecular electronics aims at attaining miniaturisation in line and/or beyond Moore's law, quantum computing applies quantum properties to information processing



Another step towards quantum computing was taken when a team of scientists processed information in the electron spin (blue) and stored it in the nuclear spin (yellow) of phosphorus atoms through a combination of microwave and radio-frequency pulses. (Image by Flavio Robles, Berkeley Lab Public Affairs)

■ **PROF. C.T. BHUNIA**

Quantum computing is a subject of nano-system that is supposed to rule the world of future computing. Just to

give you an idea of the nano world, note that 1 nanometre is a billionth of a metre and 0.1 nm is the size of the hydrogen atom.

Computing at the nano scale is required as further chip integration may

not be possible in the near future. The quantum effect stands in the way of integration in tune with Moore's law.

Moore's laws state that the number of components on an IC would double every year (this is the original Moore's law), circuit complexity on an IC would double every 18 months (this is known as revised Moore's law) and the processing power of computer would double every year-and-a-half (Moore's second law).

Already, ICs are being made of around 250 million transistors. If Moore's law continues to hold good, it is predicted that by 2010 end ICs will be made of a billion transistors. Threats to the survival of Moore's laws are heat dissipation and quantum effect, which is a physical limit to IC integration. Several predictions were therefore made for imminent death of Moore's laws. Contrary to these predictions, Moore's laws are surviving and hold true for IC integration.

Recently, two research reports have showed that Moore's laws will hold true for another few years. A survey conducted jointly by the Institute of Electrical and Electronics Engineers (IEEE) and the Response Center Inc. of USA (a market research firm) over the fellows of IEEE showed that 17 per cent, 52 per cent and 31 per cent respondents predicted continuation of Moore's laws for more than ten years, 5-10 years and less than five years, respectively. The average predicted life term for the laws is thus about six years.

The expectation of realising a billion-transistors IC by 2010 has been further brightened by the current research of Intel expanding Moore's